JavaScript Lab 5

Question 1: Tip of the Day

1. Create an array that contains **10 JavaScript tips**.
    - Each element in the array should be a **string**.
    - Each string represents one useful JavaScript tip.
2. Display a **random tip** from the array every time the page is opened.
    - This will act as a **"Tip of the Day"** feature.

Question 2: Students Grades

1. Create an array that holds the following students grades:
   `60, 100, 10, 15, 85`
2. Perform the following tasks:
   a. Use the `filter()` method to filter the passing students (grades of 50 or more).
   b. Use the `reduce()` method to get the average grade of all students.
   c. Sort the array **numerically in descending order (Use `sort()` method)**.
   d. Using the **sorted array**, find the **highest student degree** that is **less than or equal to 100**.
   e. Print the grades of all students who scored **below 60 (Use `forEach()` method)**.

Question 3: Students Data

1. Create an array of objects where:
    - Each object represents a student.
    - Each object has two properties:
        - `Name`
        - `Age`
        - `Grades array`
        - `printInfo()` function to print the all the student's properties

2. Perform the following tasks:
   a. Find the **first student name** who got a degree **between 90 and 100**.
      - Use the `find()` method.
3. b. Print the **names of students** who got a degree **less than 60** (Use the `filter()` method.).
   c. Add a **new student** to the array, then use `forEach()` to print **all students**.
   d. Remove the **last student** from the array using the `pop()` method, then use `forEach()` to print **all remaining students**.
   e. Sort the array **alphabetically** based on the student name.
   f. Use the `splice()` method to add **two new students after the second element** of the array.
   g. Use the `splice()` method to remove **one student after the third element** of the array.
   h. **Map** every student object to create a **new array** where every student object has an average grade property.

Question 4: Attendance and Performance Analyzer

You are given the following array of student objects:

```
const students = [
  {
    name: "Ahmed",
    grades: [85, 90, 78],
    attendance: [true, true, true, false, true]
  },
  {
    name: "Sara",
    grades: [60, 55, 70],
    attendance: [true, false, false, true, false]
  },
  {
    name: "Omar",
    grades: [95, 92, 88],
    attendance: [true, true, true, true, true]
  },
  {
    name: "Lina",
    grades: [45, 50, 58],
    attendance: [true, false, true, false, true]
  }
];
```

Required Tasks:

1. Calculate the attendance percentage for each student.
2. Add a new property called `attendanceRate` to each student object.
3. Find all students whose attendance rate is less than 75 percent.
4. Among students with an attendance rate of 75 percent or higher, find the student with the highest average grade.
5. Create a new array that contains only the names of students eligible for an honor badge.

Honor badge rules:

- Attendance rate must be at least 90 percent.
- Average grade must be at least 85.

Question 5: Course Enrollment and Data Transformation

You are given the following array of course objects:

```
const courses = [
  {
    courseName: "JavaScript",
    instructor: "Mr. Smith",
    students: [
      { name: "Ahmed", grades: [80, 90, 85] },
      { name: "Sara", grades: [60, 65, 70] },
      { name: "Omar", grades: [95, 92, 88] }
    ]
  },
  {
    courseName: "Databases",
    instructor: "Ms. Johnson",
    students: [
      { name: "Lina", grades: [55, 58, 60] },
      { name: "Yousef", grades: [72, 75, 78] },
      { name: "Mona", grades: [88, 90, 92] }
    ]
  }
];
```

Required Tasks:

1. Calculate the average grade for each student inside each course.
2. Add a new property called `averageGrade` to every student object.
3. Calculate the overall course average for each course using the students' average grades.
4. Create a new array that contains only courses with a course average greater than 70.
5. From those courses, create a single flat array of student names whose average grade is below their course average.
6. Sort the final array of student names alphabetically.
